Forme standard

$$\text{Min } Z = -5x_1 - 4x_2$$

$$\text{sc: } \begin{cases} x_1 + x_2 + e_1 = 5 \\ 2x_1 + 3x_2 - u_1 + w_1 = 12 \\ x_1 + 4x_2 - u_2 + w_2 = 8 \end{cases}$$

$(x_1 ; x_2 ; e_1 ; u_1 ; u_2 ; w_1 ; w_2 \geq 0)$

Tableau du simplexe 1o :

	x_1	x_2	e_1	u_1	u_2	w_1	w_2	C
e_1	1	1	1	0	0	0	0	5
w_1	2	3	0	-1	0	1	0	12
w_2	1	4	0	0	-1	0	1	8
Z	5	4	0	0	0	0	0	Z=0.

$Z = Z_j - C_j$

$x_1 = 0$
$x_2 = 0$
dans la BI
$\Leftrightarrow Z = 0.$

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Phase I : Changement d'objectif :

$\text{Min } Z_A = \omega_1 + \omega_2 \quad (Z_A = 0)$

$\omega_1 = 12 - 2x_1 - 3x_2 + u_1$

$\omega_2 = 8 - x_1 - 4x_2 + u_2$

$\text{Min } Z_A = 20 - 3x_1 - 7x_2 + u_1 + u_2$

To1		x_1	x_2 (Vé)	e_1	u_1	u_2	ω_1	ω_2	C	K
0	e_1	1	1	1	0	0	0	0	5	5
0	ω_1	2	3	0	-1	0	1	0	12	4
0	ω_2 (VP)	1	(4)	0	0	-1	0	1	8	2
	Z_A	3	7	0	-1	-1	0	0	$Z_A = 20$	

To2		x_1 (Vé)	x_2	e_1	u_1	u_2	ω_1	ω_2	C	K
e_1	(VP)	3/4	0	1	0	1/4	0	-1/4	3	4
ω_1		5/4	0	0	-1	3/4	1	-3/4	6	4,8
x_2		1/4	1*	0	0	-1/4	0	1/4	2	8
	Z_A	5/4	0	0	-1	3/4	0	-7/4	6.	

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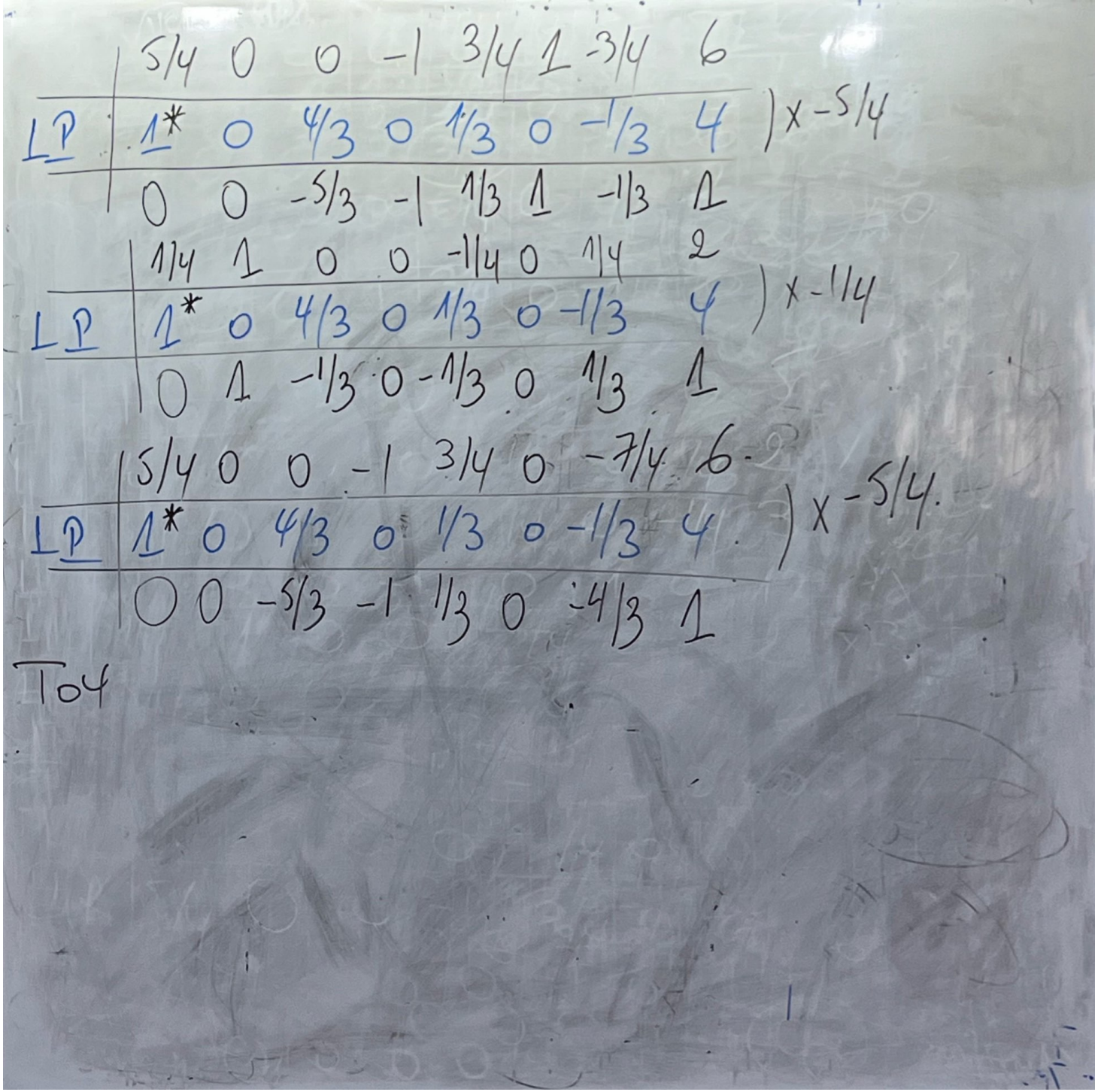
LC	1	(1)	1	0	0	0	0	5
LP	1/4	1*	0	0	-1/4	0	1/4	2) x-1
LR	3/4	0	1	0	1/4	0	-1/4	3.

LC	2	(3)	0	-1	0	1	0	12
LP	1/4	1*	0	0	-1/4	0	1/4	2) x-3
LR	5/4	0	0	-1	3/4	1	-3/4	6

LC	3	(7)	0	-1	-1	0	0	20
LP	1/4	1*	0	0	-1/4	0	1/4	2) x-7
LR	5/4	0	0	-1	3/4	0	-7/4	6.

103	x_1	x_2	e_1	u_1	u_2	w_1	w_2	C
x_1	1*	0	4/3	0	1/3	0	-1/3	4
w_1	0	0	-5/3	-1	1/3	1	-1/3	1
x_2	0	1	-1/3	0	-1/3	0	1/3	1
Z_A	0	0	-5/3	-1	1/3	0	-4/3	1

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	5/4	0	0	-1	3/4	1	-3/4	6
LP	1*	0	4/3	0	1/3	0	-1/3	4) x -5/4
	0	0	-5/3	-1	1/3	1	-1/3	1

	1/4	1	0	0	-1/4	0	1/4	2
LP	1*	0	4/3	0	1/3	0	-1/3	4) x -1/4
	0	1	-1/3	0	-1/3	0	1/3	1

	5/4	0	0	-1	3/4	0	-7/4	6
LP	1*	0	4/3	0	1/3	0	-1/3	4) x -5/4
	0	0	-5/3	-1	1/3	0	-4/3	1

Tot	x_1	x_2	e_1	u_1	u_2	~~u_3~~	~~u_4~~	C
x_1	1	0	3	1	0	-1	0	3
u_2	0	0	-5	-3	1*	3	-1	3
x_2	0	1	-2	-1	0	1	0	2
Z_A	0	0	0	0	0	-1	-1	0

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LC	1	0	4/3	0	1/3	0	-1/3	4
LP	0	0	-5	-3	1*	3	-1	3
LR	1	0	3	1	0	-1	0	3

) x -1/3.

LC	0	1	-1/3	0	(-1/3)	0	1/3	1
LP	0	0	-5	-3	1*	3	-1	3
LR	0	1	-2	-1	0	1	0	2

) x 1/3

LC	0	0	-5/3	-1	1/3	0	-4/3	1
LP	0	0	-5	-3	1*	3	-1	3
LR	0	0	0	0	0	-1	-1	0

) ÷ 1/3

Phase II : Min Z = $-5x_1 - 4x_2$

		x_1	x_2	e_i	u_1	u_2	C
-5	x_1	1	0	3	1	0	3.
0	u_2	0	0	-5	-3	1	3.
-4	x_2	0	1	-2	-1	0	2
	Z	0	0	(-7)	(-1)	0	(-23)

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	5/4	0	0	-1	3/4	1	-3/4	6
LP	1*	0	4/3	0	1/3	0	-1/3	4) x -5/4
	0	0	-5/3	-1	1/3	1	-1/3	1

	1/4	1	0	0	-1/4	0	1/4	2
LP	1*	0	4/3	0	1/3	0	-1/3	4) x -1/4
	0	1	-1/3	0	-1/3	0	1/3	1

	5/4	0	0	-1	3/4	0	-7/4	6
LP	1*	0	4/3	0	1/3	0	-1/3	4) x -5/4
	0	0	-5/3	-1	1/3	0	-4/3	1

Tot	x_1	x_2	e_1	u_1	u_2	~~u_3~~	~~u_4~~	C
x_1	1	0	3	1	0	-1	0	3
u_2	0	0	-5	-3	1*	3	-1	3
x_2	0	1	-2	-1	0	1	0	2
Z_A	0	0	0	0	0	-1	-1	0

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LC	1	0	4/3	0	1/3	0	-1/3	4
LP	0	0	-5	-3	1*	3	-1	3
LR	1	0	3	1	0	-1	0	3

) x -1/3.

LC	0	1	-1/3	0	(-1/3)	0	1/3	1
LP	0	0	-5	-3	1*	3	-1	3
LR	0	1	-2	-1	0	1	0	2

) x 1/3

LC	0	0	-5/3	-1	1/3	0	-4/3	1
LP	0	0	-5	-3	1*	3	-1	3
LR	0	0	0	0	0	-1	-1	0

) ÷ 1/3

Phase II : Min Z = $-5x_1 - 4x_2$

		x_1	x_2	e_i	u_1	u_2	C
-5	x_1	1	0	3	1	0	3.
0	u_2	0	0	-5	-3	1	3.
-4	x_2	0	1	-2	-1	0	2
	Z	0	0	(-7)	(-1)	0	(-23)

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